KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

# MATHEMATICS

## Year 8 – Yearly Examination

## Equations, Investigating Data and Mixed Topics

| **Instructions** | | |
|---|---|---|
| 1. Attempt all questions<br>2. Show working where needed<br>3. Time Allowed 75 minutes<br>4. Calculators are permitted<br>5. Marks are shown beside each question | Equations (Section A): | /20 |
| | Investigating Data (Section B): | /23 |
| | Mixed Topics (Section C): | /22 |
| | Total Marks: | /65 |

## Section A – Equations (20 marks)

*Suggested time: 25 minutes. Show all working.*

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**Question 1 – One-step equations** **(4 marks)**

Solve each equation.

a) x + 7 = 15 x = _____ **(1 mark)**

Working: _____

b) m − 9 = 4 m = _____ **(1 mark)**

Working: _____

c) 6p = 42 p = _____ **(1 mark)**

Working: _____

d) y ÷ 5 = 3 y = _____ **(1 mark)**

Working: _____

**Question 2 – Two-step equations** **(6 marks)**

Solve each equation, showing your working.

a) 3x + 5 = 20 **(2 marks)**

Working: _____

_____

_____

x = _____

b) 4y − 7 = 21 **(2 marks)**

Working: _____

_____

_____

y = _____

c) x ÷ 3 + 2 = 9 **(2 marks)**

Working: _____

_____

_____

x = _____

**Question 3 – Equations with a negative solution** **(2 marks)**

Solve 2m + 9 = 3.

Working: _____

_____

_____

m = _____

**Question 4 – Checking solutions** **(2 marks)**

Use substitution to show whether $x = 4$ is the solution to $3x - 5 = 7$.

Working: _____

_____

_____

_____

Is $x = 4$ the solution? (circle one) YES / NO

**Question 5 – Worded problem** **(3 marks)**

I think of a number, multiply it by 4, and then subtract 6. The answer is 22.

a) Write an equation to represent this problem. Use n for the number. **(1 mark)**

Equation: _____

b) Solve your equation to find the number. **(2 marks)**

Working: _____

_____

_____

n = _____

**Question 6 – Worded problem** **(3 marks)**

Mia buys 5 pens that each cost the same amount and a ruler that costs $3. She spends $18 altogether. Let p be the cost of one pen in dollars.

a) Write an equation to represent this situation. **(1 mark)**

Equation: _____

b) Solve your equation to find the cost of one pen. **(2 marks)**

Working: _____

_____

_____

p = _____

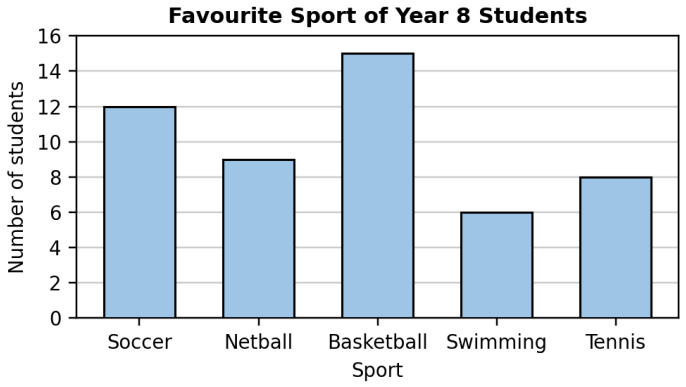
## Section B – Investigating Data (23 marks)

*Suggested time: 25 minutes. Show all working.*

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**Question 7 – Column graph** **(2 marks)**

Year 8 students were surveyed about their favourite sport. The results are shown below.



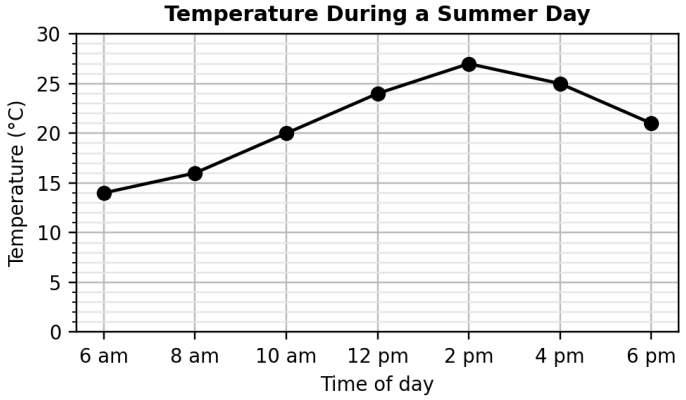

a) Which sport was the most popular? _____ **(1 mark)**

b) How many students were surveyed altogether? _____ **(1 mark)**

Working: _____

**Question 8 – Line graph** **(2 marks)**

The line graph shows the temperature at different times on a summer day.




a) At what time was the temperature the highest? _____ **(1 mark)**

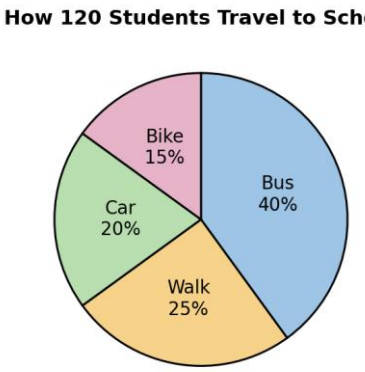
b) By how many degrees did the temperature rise between 6 am and 12 pm? **(1 mark)**

Working: _____

Answer: _____

**Question 9 – Sector (pie) graph** **(3 marks)**

120 students were asked how they usually travel to school. The results are shown in the sector graph.




a) How many students travel to school by bus? **(1 mark)**

Working: _____

Answer: _____

b) How many more students walk than ride a bike? **(2 marks)**

Working: _____

_____

_____

Answer: _____

**Question 10 – Stem-and-leaf plot** **(3 marks)**

The test scores (out of 60) of 15 students are shown in the stem-and-leaf plot.

| Stem | Leaf |
|---|---|
| **2** | 3 7 9 |
| **3** | 1 4 4 8 |
| **4** | 0 2 5 6 7 9 |
| **5** | 1 3 |

*Key: 2 | 3 means 23*

a) How many students sat the test? _____ **(1 mark)**

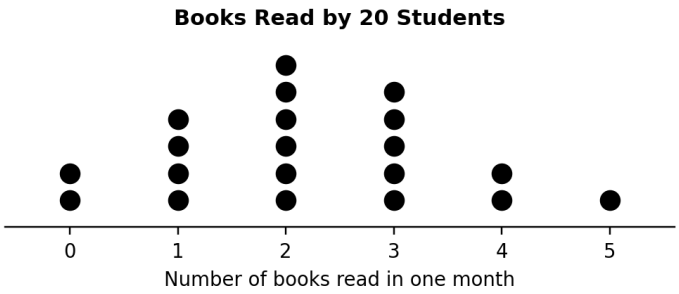
b) Find the median score. _____ **(1 mark)**

Working: _____

c) How many students scored more than 40? _____ **(1 mark)**

**Question 11 – Dot plot** **(2 marks)**

Twenty students were asked how many books they read last month. The results are shown in the dot plot.




a) What is the mode? _____ **(1 mark)**

b) How many students read at least 4 books? _____ **(1 mark)**

**Question 12 – Frequency table** **(3 marks)**

Students in Year 8 were asked how many pets they own. The results are shown in the frequency table.

| **Number of pets** | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| **Frequency** | 4 | 9 | 10 | 5 | 2 |

a) How many students were surveyed altogether? _____ **(1 mark)**

Working: _____

b) What is the mode? _____ **(1 mark)**

c) How many students own at least 2 pets? _____ **(1 mark)**

**Question 13 – Mean, median, mode and range** **(5 marks)**

The number of minutes that eight students spent on homework last night is shown below.

**14, 9, 11, 14, 8, 16, 14, 10**

a) Calculate the mean. **(2 marks)**

Working: _____

_____

_____

Mean = _____

b) Find the median. _____ **(1 mark)**

Working: _____

c) Find the mode. _____ **(1 mark)**

d) Find the range. _____ **(1 mark)**

Working: _____

**Question 14 – Outliers** **(3 marks)**

The number of text messages sent in one day by six students is shown below.

**12, 55, 14, 11, 13, 15**

a) Identify the outlier in this data set. _____ **(1 mark)**

b) Calculate the mean of the data, including the outlier. **(2 marks)**

Working: _____

_____

_____

Mean = _____

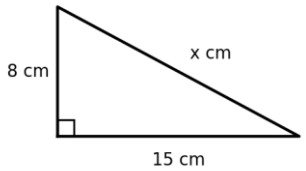
## Section C – Mixed Topics (22 marks)

*Suggested time: 25 minutes. Show all working.*

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**Question 15 – Pythagoras' Theorem** **(4 marks)**

a) Find the length of the hypotenuse, x, in the triangle below. **(2 marks)**




Working: _____

_____

_____

x = _____

b) A triangle has side lengths 7 cm, 24 cm and 25 cm. Is it a right-angled triangle? Show working. **(2 marks)**

Working: _____

_____

_____

_____

**Question 16 – Decimals** **(2 marks)**

a) Calculate 4.8 × 0.35 _____ **(1 mark)**

Working: _____

_____

b) Calculate 7.2 ÷ 0.06 _____ **(1 mark)**

Working: _____

_____

**Question 17 – Powers and roots** **(2 marks)**

a) Evaluate $2^4$ _____ **(1 mark)**

Working: _____

b) Evaluate $\sqrt{64}$ _____ **(1 mark)**

Working: _____

**Question 18 – Surds** **(3 marks)**

a) Circle the surds: $\sqrt{9}$ $\sqrt{12}$ $\sqrt{25}$ $\sqrt{11}$ **(1 mark)**

b) Simplify $\sqrt{50}$ **(2 marks)**

Working: _____

_____

_____

Answer: _____

**Question 19 – Algebra** **(3 marks)**

a) Simplify $4a + 3b - a + 5b$ _____ **(1 mark)**

Working: _____

b) Expand $3(x + 4)$ _____ **(1 mark)**

Working: _____

c) Evaluate $2m^2 - 3n$ when $m = 4$ and $n = 5$. _____ **(1 mark)**

Working: _____

_____

**Question 20 – Area and volume** **(4 marks)**

a) Find the area of a trapezium with parallel sides 8 cm and 14 cm and height 5 cm. **(2 marks)**

*Formula: Area = ½ × (a + b) × h, where a and b are the parallel sides and h is the perpendicular height.*

Working: _____

_____

_____

Answer: _____

b) Find the volume of a rectangular prism that is 9 cm long, 4 cm wide and 3 cm high. **(2 marks)**

Working: _____

_____

_____

Answer: _____

**Question 21 – Linear relationships** **(2 marks)**

Complete the table of values for the rule $y = 2x - 3$.

| **x** | −1 | 0 | 1 | 2 |
|---|---|---|---|---|
| **y** | | | | |

**Question 22 – Rates and ratios** **(2 marks)**

a) Write the ratio 18 : 24 in simplest form. _____ **(1 mark)**

Working: _____

_____

b) A car travels 210 km in 3 hours. Find its average speed in km/h. _____ **(1 mark)**

Working: _____

_____

**END OF EXAMINATION PAPER**